

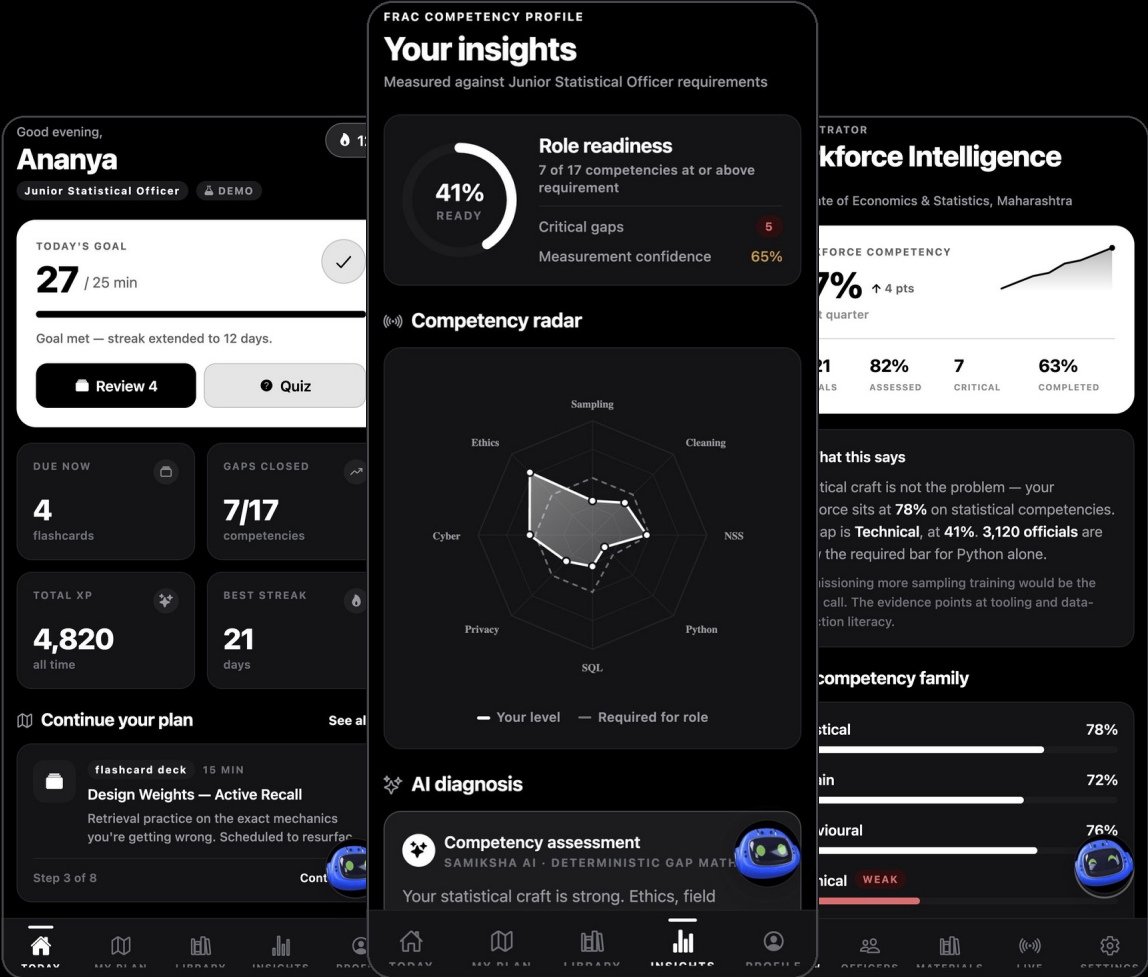
समीक्षा

Samiksha

AI-enabled capacity building for India's Official Statistical System

Measures every officer against the FRAC competencies their post requires, closes the gaps through iGOT Karmayogi and grounded practice from department material — and gives them AI agents that know where they are weak.

- Officer app
- Admin app
- AI Workspace
- Samiksha AI

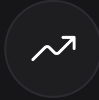


Capacity building that cannot see the gap

SIH26101 · MoSPI

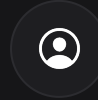
“Develop an AI enabled learning platform that identifies competency gaps, recommends personalized training through integration with the iGOT Karmayogi ecosystem, and capable of generating Quizzes and MCQs from uploaded learning materials to strengthen capacity building in India's Official Statistical System.”

Thousands of officers across MoSPI, NSO, NSSTA and every state DES produce the figures behind budgets, monetary policy and welfare.



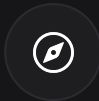
Skills are shifting

Python, SQL, GIS and the DPDP Act are now expected alongside sampling and estimation.



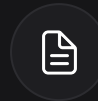
Gaps are self-reported

Nominations and forms — and people systematically misjudge their own skill.



Training is generic

Large catalogues, but no order and no link to an individual's gap.



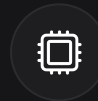
MCQs are hand-written

Few assessments exist; handbooks and circulars are read once and forgotten.



No workforce view

Enrolment counts, not where a directorate is collectively weak.



AI does work, not learning

Productivity rises while capability — and dependency — stay invisible.

The statement hides **two AI engines × two audiences** — most solutions build only half.

Two halves that feed each other — the loop is the point

LEARN · MOBILE APP

Measure & teach

- Role → FRAC competency requirements
- Adaptive test & quizzes → evidence
- SQL gap arithmetic with confidence
- Prerequisite-aware plan via iGOT Karmayogi
- Quizzes, flashcards, mock tests from material
- Misconceptions, calibration, AI tutor

THE COMPETENCY LOOP

Measured gaps → agent prompt



*"You delegated 23 Python tasks this month
— closing this gap gives that work back."*



Delegations by competency →
evidence

DO · AI WORKSPACE

Delegate & do

- Data Quality · Data Analyst · Report Assistant
- Own computer: browser, terminal, files, desktop
- Same sign-in as the mobile app
- Told the officer's gaps before every run
- Explains the step that matters
- Every run ends at a draft — people decide



Samiksha

Officer app · Android



Samiksha Admin

Administrator app · Android



Demo apps

Both journeys, no account

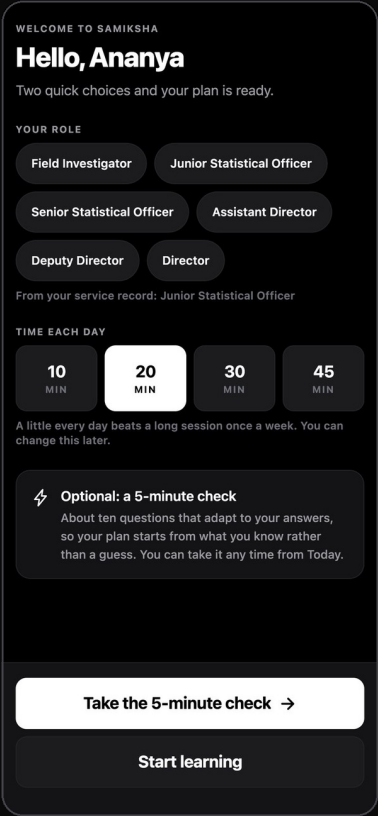


Samiksha Workspace

AI agents · Windows

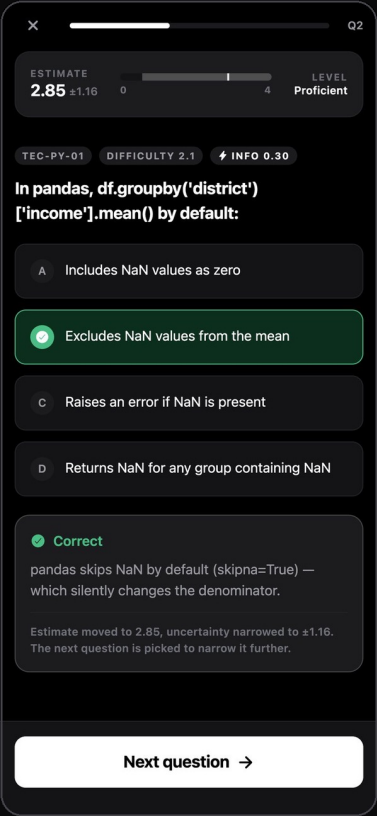
Principles Evidence over opinion · Deterministic core, AI at the edges · Agents draft, people decide · One identity everywhere

Measured, planned, practised — in minutes a day



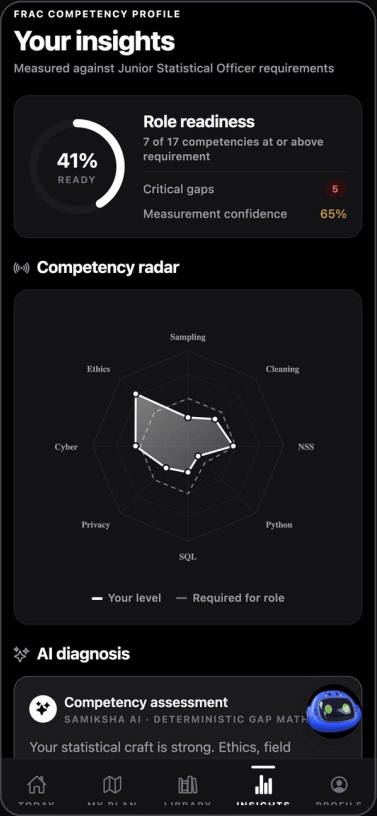
1 Two-tap onboarding

Account provisioned against an employee ID; confirm role & daily goal.



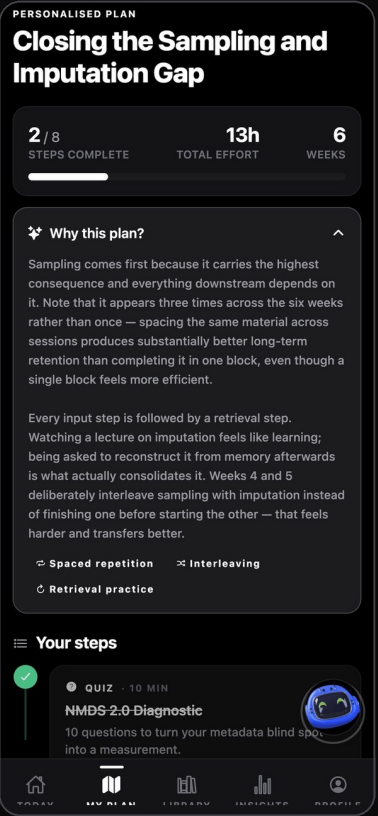
2 Adaptive diagnostic

IRT picks the most informative question; uncertainty band narrows live.



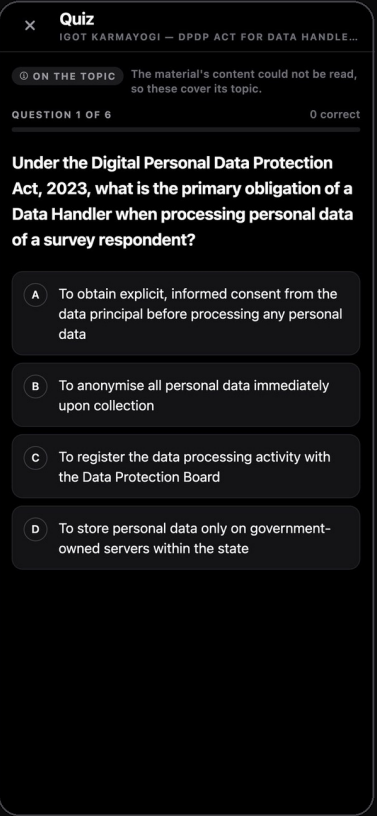
3 Competency profile

Readiness, radar vs role requirement, AI diagnosis, misconceptions.



4 Personalised plan

iGOT courses sequenced by prerequisites, each step with its reason.



5 Practise on material

Quiz, flashcards or mock test written from department material.

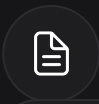
Trustworthy by construction: the maths is not the model's



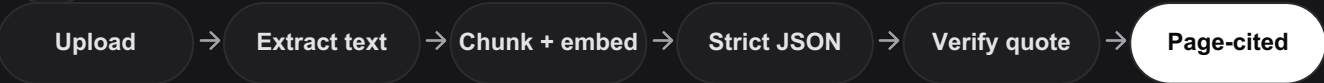
Gap analysis is SQL — AI only explains it

```
priority = max(0, required - score) × weight × criticality × (0.5 + 0.5 · confidence)
```

Reproducible and auditable. Low confidence means “assess first”, not “train” — no budget spent on unmeasured skills.



Grounded MCQs from any upload



Every question quotes its source verbatim; the quote is checked in the document and mapped to a page or slide. Each distractor names the misconception it encodes.

IRT adaptive test
≈ half the questions

FSRS-6 review
matches reference intervals

Prerequisite graph
learnable next, not biggest gap

Misconception tracking
which wrong answer, not just wrong

Confidence calibration
overconfidence named

Bloom spread
apply · analyse · evaluate

1/5

ignoring that households were also sub-sampled.

✓

The product of the inverses of both stage-one and stage-two selection probabilities

C

The total district population divided by 320

ⓘ

Confuses a raw expansion factor with a design weight; PPS selection means villages did not have equal probability.

D

The inverse of the within-village selection probability only

ⓘ

Ignores the PPS village stage entirely — the most common error when only the final stage feels like 'the sampling'.

✓ Correct

In multi-stage sampling the overall selection probability is the product of the stage probabilities, so the design weight is the product of their inverses: $w = 1/(P_{\text{village}} \times P_{\text{household|village}})$. Using only one stage discards the other stage's unequal selection and biases the estimate.

✓ Grounded in your material

Next question →

Feedback: rationale & source

AI diagnosis

Competency assessment

SAMIKSHA AI · DETERMINISTIC GAP MATHS

Your statistical craft is strong. Ethics, field operations and disclosure control all sit at or above what a Junior Statistical Officer post requires, and that is the half of the job most people find hardest to build.

The gap is not statistical — it is tooling and compliance. You are at 0.6/4 on Python and 1.1/4 on SQL, which means extraction and cleaning work that should take an afternoon is taking you days, and you are dependent on someone else for every non-trivial data pull. Alongside that, DPDP Act questions are going wrong often enough to be a compliance exposure now that the Act is in force.

Sampling remains the highest-consequence statistical gap: you handle the vocabulary confidently, which masks that multi-stage weight computation is going wrong. GIS looks low but has only three assessed items — that is a measurement gap, not a skill gap, and should be tested before anything is prescribed.

STRENGTHS

✓ BEH-INT-01

Consistently correct on confidentiality obligations and on scenarios involving pressure to alter a release. This is the competency that protects the institution, and you are above the bar for your grade.

✓ FUN-COLL-01

Near-proficient in field operations with strong

HOME

MY PLAN

LIBRARY

INSIGHTS

PROFILE

AI diagnosis of computed gaps

See the workforce, not just enrolments



Workforce Intelligence

Where the directorate is weak, how many officials are below the bar, quarterly trend — aggregate by construction.



Roster & officer record

Presence, ranked gaps, repeated misconceptions and the prescribed next step on one screen.



Publish study material

PDF, slides, docs, video or links to a department or named officers; open rates tracked.



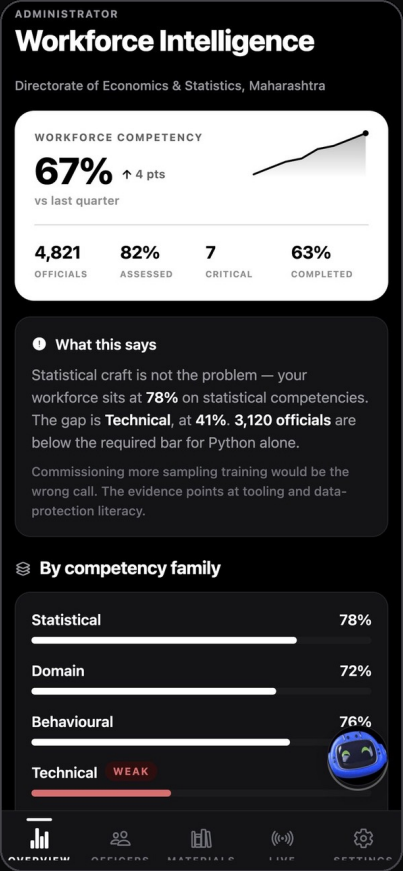
Samiksha AI operator

Asks for live data; publishing or account changes come back as actions to confirm.

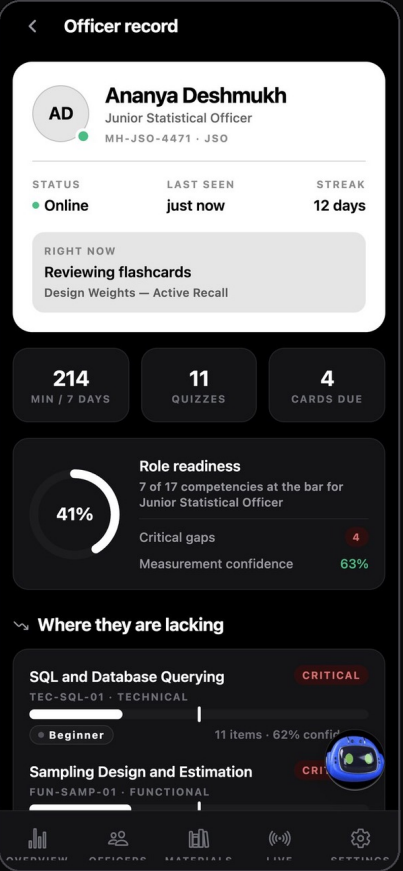


Provisioned accounts only

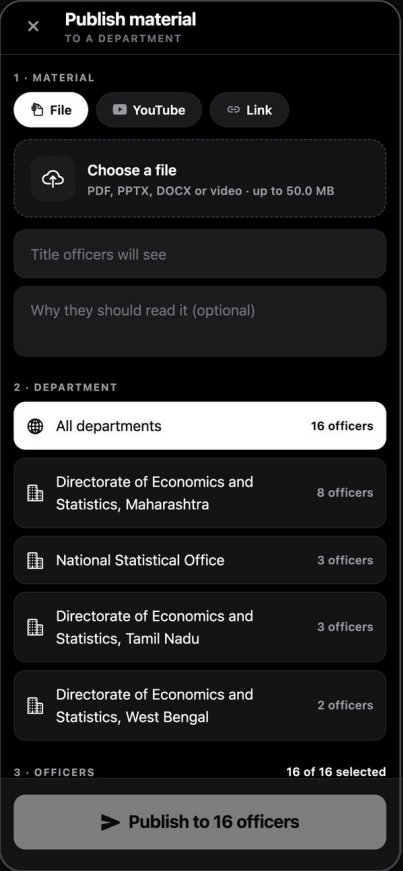
Created against verified employee IDs; reset, deactivate, reactivate.



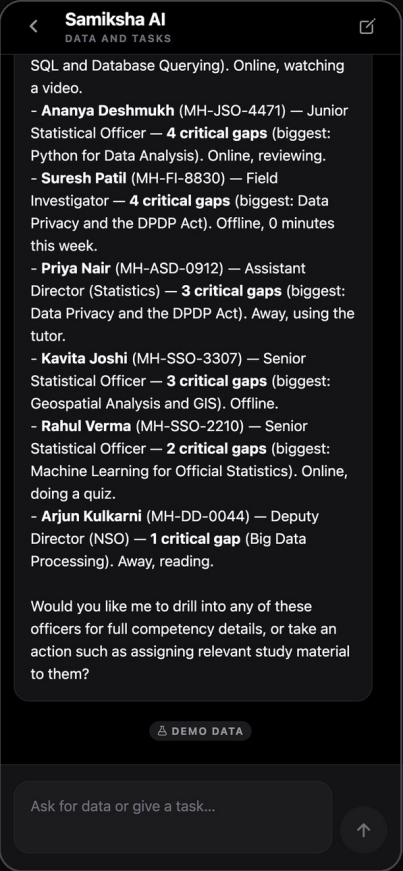
Workforce Intelligence



Officer record



Publish to officers



Samiksha AI · live data

Delegation becomes evidence about competency



Data Quality

FUN-CLEAN-01 · FUN-QUAL-01

Missing-data patterns, duplicates, units, implausible values. Proposes fixes — never repairs silently.



Data Analyst

TEC-PY-01 · FUN-ANAL-01 · FUN-SAMP-01 · TEC-SQL-01

Applies design weights, computes indicators and tables; asks before assuming SRS.



Report Assistant

FUN-META-01 · FUN-DISS-01 · BEH-COM-01

Drafts against approved templates; every figure traceable; never sends onward.

EVERY RUN

1

Same sign-in

Credentials proved against Supabase; one user ID

→

2

Gaps loaded

Post, gaps & strengths into the system prompt

→

3

Work + teaching

Explains the step where the officer is weak

→

4

Delegation logged

Run recorded against the agent's competency

`v_officer_dependency` joins delegation volume to the measured gap. Delegating what you can already do scores zero — that is good delegation. Delegating what your post needs you to do is shown to the officer as “What you are handing to your agents”.

FUN-CLEAN-01

3x · 4 DAYS AGO

Believes preserving the mean means the imputation was harmless — does not consider what happens to the variance.

FUN-SAMP-01

2x · 1 WEEK AGO

Confuses a raw expansion factor with a design weight under unequal selection probability.

Target these with the tutor

What you are handing to your agents

Delegating work is the point of having agents. These are the ones where you are also **below the level your post requires** — so the agent is covering a gap rather than saving you time.

Python for Data Analysis

23

TEC-PY-01 · VIA DATA ANALYST · LAST 30 DAYS

Every dataset you have been given this month went to the agent. The work you get back is correct, but the analysis is not yours yet.

8H SAVED · STILL A GAP

Data Cleaning, Editing and I...

14

FUN-CLEAN-01 · VIA DATA QUALITY · LAST 30 DAYS

You catch the missing values once the agent flags them. Reading the pattern yourself is the step still missing.

4H SAVED · STILL A GAP

Metadata and Statistical Sta...

6

TODAY

MY PLAN

LIBRARY

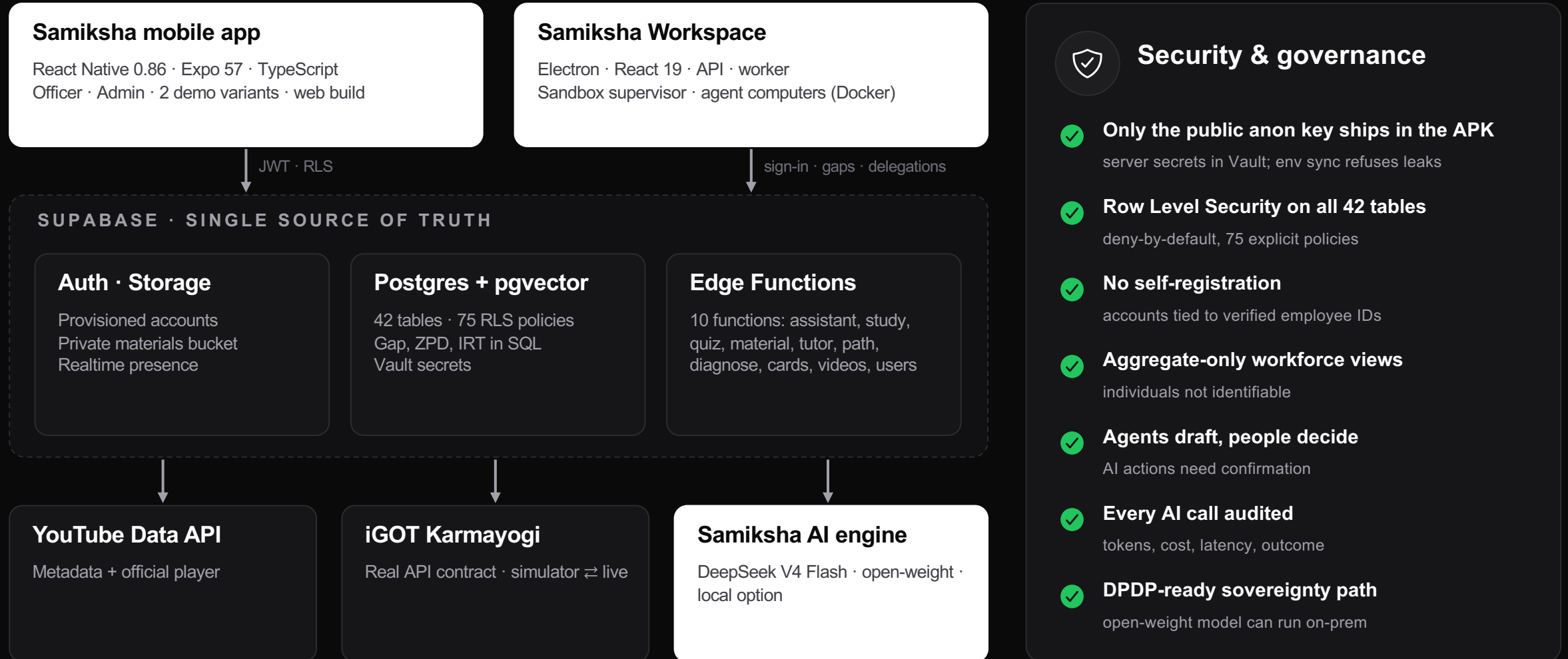
INSIGHTS

PROFILE

Samiksha · SIH26101

07

One source of truth, no secrets on the device



Why DeepSeek V4 Flash: open weights, built to be fine-tuned



Easy to fine-tune

Open weights under a permissive licence — LoRA / QLoRA adapters on MoSPI & NSSTA manuals, SQAF, NMDS and trainer-approved questions. Closed API models can't be adapted this way.



Sovereign hosting

The same model, base or fine-tuned, can run on NIC / MeghRaj infrastructure — officer data never leaves India (DPDP Act, 2023).



~1M-token context

A whole 150-page handbook in one call — no chunk-stitching, no contradictory questions across chunks.



Cost at workforce scale

≈ US\$0.09 per million input tokens: a 20-question quiz from a full PDF costs under a cent. Efficient mixture-of-experts design.



Strict JSON & tool calling

Grammar-constrained output guarantees shape; tools power the administrator operator.



No lock-in

One gateway, one env var. A local model path through Ollama is already proven for offline / on-prem.

FINE-TUNING ROADMAP

The training data is already being collected

- 1 **Collect**
Audit log · grounded questions · misconceptions · edits
- 2 **Adapt**
LoRA on the public MoSPI / NSSTA corpus
- 3 **Supervise**
SFT on trainer-approved quizzes, diagnoses, plans
- 4 **Align**
Preference tuning from trainer corrections
- 5 **Evaluate**
Grounding rate · schema validity · calibrated difficulty
- 6 **Deploy**
Government cloud, same gateway, zero app changes

Running today, honest about what comes next

39

FRAC competencies
across 4 families

6

posts, Field Investigator
to Director

42

tables under
Row Level Security

10

server-side AI &
admin functions

5

installable apps
per release

Built & verified

- ✓ Identity bridge — one login, phone & desktop
- ✓ Competency loop in both directions
- ✓ Adaptive assessment · FSRs review
- ✓ Grounded quizzes with page citations
- ✓ AI practice from any department material
- ✓ Samiksha AI for officers & administrators
- ✓ Both dashboards · publishing · live presence
- ✓ Signed APKs + Windows installer per release

Stated plainly

I G O T

Real contract, simulator → go live

F R A C

Reconstructed → official import

M O D E L

Base weights → fine-tuned

L A N G U A G E

English UI → Hindi & regional

Roadmap

Pilot

One state DES · live iGOT · trainer review of questions

Scale

Fine-tuned model on government cloud · languages · offline sync

Ecosystem

Karmayogi passbook sync · training-impact evaluation · CPI / ASI / PLFS agent templates

“The measure of success is not how much work the AI does, but how much more capable each officer becomes.”

github.com/adryxportfolio/sih2026
Releases · documentation · source